Retriever-Conditioned Representation Search for Cross-Domain Patent-Family Retrieval

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Abstract

Cross-domain prior-art retrieval must recover relevant patent families even when the query and target use different technical language. Representation is usually fixed before retriever evaluation, although field selection, segmentation, packing, and family aggregation determine what a retriever can score. We test whether representation should instead be selected conditional on a frozen retriever. Retriever-Conditioned Representation Search (RCRS) evaluates a finite grammar of executable patent-text programs without updating retriever weights. Five lexical or dense retrievers were screened on a family-level cross-domain benchmark. Development results varied by retriever: program transfer was asymmetric, one search surface collapsed to a single effective action, and fusing every primary system did not beat the strongest single system. One 125-query selection exposure fixed two systems for an 872-query confirmation. RCRS increased OUT Recall@100 from 0.3311 to 0.4425 (paired difference 0.1114; 95% bootstrap confidence interval 0.1023–0.1204), with 619 wins, 158 ties, and 95 losses. The frozen winner was then run on the complete 1,247-query benchmark. Of 5,193 strict OUT-relevant family pairs, 796 appeared in the first 100 ranks, 332 only in ranks 101–200, and 4,065 were absent from the Top-200 pool. Conditional representation search therefore produced a confirmed gain under the frozen protocol, while the post-confirmatory diagnosis placed most remaining strict OUT error outside the exposed candidate pool.

Keywords: patent retrieval, patent families, representation search, cross-domain retrieval, candidate exposure, reproducible evaluation

1. Introduction

A patent retrieval system rarely scores an invention as a single, uniform document. Technical content may be split across family members and across titles, abstracts, descriptions, and claims. Claims compress legal scope into specialized language; descriptions repeat concepts at several levels of detail. Cross-domain retrieval adds vocabulary shift between the query and the relevant family. These properties make text construction part of the retrieval configuration. Field order, structural labels, passage boundaries, packing, duplicate handling, and family aggregation all affect the evidence presented to a frozen retriever.

Family-level and embedding benchmarks now support controlled study of these choices. DAPFAM defines domain-aware relevance at patent-family level [1]. PatenTEB and a broader multi-task study compare patent embeddings across retrieval, classification, and clustering tasks [2, 3]. Those studies primarily compare models or embeddings. Here the model weights stay fixed while the executable representation changes.

AutoIndex frames representation construction as a program-search problem [4]. Patent records require a more constrained grammar than generic text: the program must respect document structure, long fields, family duplication, and candidate-level aggregation. Optimization also needs an evaluation protocol that separates iterative development from confirmation. Repeated access to one final set would make a representation program another route for test-set adaptation.

Retriever-Conditioned Representation Search (RCRS) addresses both points. The experimental repository records the implementation as ArmIndex; RCRS is the publication-facing name. The method searches serializable programs that transform patent text and aggregate family evidence. Retriever weights, training data, and relevance judgments do not change. The protocol separates iterative development, one Selection-125 exposure, paired Final-872 confirmation, and post-confirmatory analysis of the unchanged winner (Fig. 1).

Repository receipts use A1–A7 for study records; retriever identifiers remain in the provenance files. The article therefore uses step names and model names in the narrative and figures; Table 2 maps the study steps back to their repository records.

The study answers three questions. First, does one representation program transfer across frozen retrievers, or are useful choices retriever-specific?

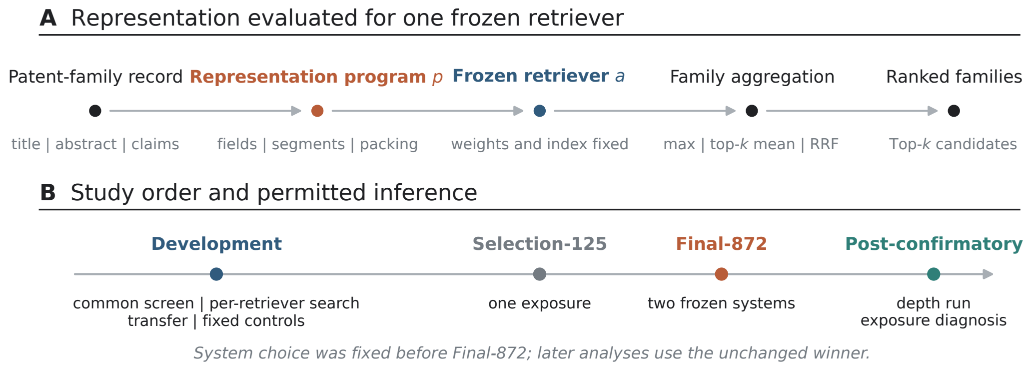

Figure 1: Method and study order. (A) An executable program constructs patent text and family-level evidence for one frozen retriever. (B) Development, Selection-125, Final-872 confirmation, and post-confirmatory analysis have separate inferential roles. System choice was fixed before Final-872.

Second, does the selected program improve a frozen system against a frozen static comparator on held-out confirmation? Third, after confirmation, how much strict OUT error reflects ordering inside the retrieved pool and how much reflects absent candidates? The claims remain within this family-level benchmark. We do not present a new embedding model, a component-level causal ablation, a universal baseline ranking, or a numerical comparison with external protocols that use different corpora or relevance definitions.

2. Related work

2.1. Patent-family retrieval and embedding evaluation

Patent retrieval differs from short-text web search in length, structure, duplication, and the cost of missed prior art. BM25 remains a useful lexical reference because it exposes exact-term evidence and has a well understood probabilistic basis [5]. Dense retrievers add semantic matching but impose model-specific context, pooling, and input conventions. Multilingual and patent-specialized embedding models therefore need evaluation at the same unit used by the application.

DAPFAM makes the family the evaluation unit and distinguishes cross-domain relations [1]. That design avoids counting several publications of the same invention as independent successes. PatenTEB and patent-embedding benchmarks cover a broader set of tasks and models [2, 3]. Their published scores cannot be placed in a common ranking with the present results unless corpus version, query construction, family normalization, relevance unit, and metric denominators match. They position the research question but do not serve as numerical comparators.

2.2. Executable representations, fusion, and candidate depth

An executable representation program records more than a prompt or a text template. It specifies the operations that produce indexable units and the rule that maps unit scores back to a patent family. AutoIndex provides the closest general formulation [4]. Our grammar specializes that idea to structured patent fields and ties every program to a frozen retriever.

Optimization in a compound system can produce flat or unstable search surfaces when distinct proposals compile to the same action or when a local change is incompatible with the surrounding components [6]. The evaluation therefore records the effective program signature and negative outcomes rather than counting proposals as independent progress. Rank fusion supplies a separate test of complementarity. Reciprocal rank fusion (RRF) combines lists without calibrated scores [7], but an extra list can still add noise. Best-single, top-two, top-three, all-primary, and commercial-only controls were fixed before evaluation.

Candidate depth separates first-stage exposure from later ordering. Passage retrieval can support fine-grained patent novelty analysis [8], but a downstream reranker or novelty model cannot act on an absent family. The post-confirmatory oracle therefore asks only what perfect ordering of the already frozen Top-200 could achieve. It is an analytical bound, not a reranker result.

3. Method

3.1. Retriever-conditioned objective

Let a denote a frozen retriever, $p \in \mathcal{P}$ an executable representation program, q a query patent record, and d a candidate patent-family record. Program p emits a finite set of candidate views,

$$\mathcal{V}_p(d) = \{v_1, \ldots, v_m\}. \tag{1}$$

The frozen retriever scores each view, and the program maps those scores to a family score,

$$S_{a,p}(q,d) = G_p(s_a(q,v_1),\ldots,s_a(q,v_m))\,. \tag{2}$$

The development search selects

$$p_a^* = \arg\max_{p\in\mathcal{P}} J_a(p), \tag{3}$$

where J_a is the prespecified family-level objective for retriever a. OUT Recall@100 is primary; latency, cost, coverage, failures, and determinism are guardrails. The subscript a is substantive: a program selected for one retriever is not assumed to transfer to another.

3.2. Executable grammar and program identity

The grammar contains auditable text operations. A program selects title, abstract, and claims; orders fields; adds structural labels; creates document, section, claim, passage, or multiview units; sets passage size, overlap, and boundary handling; packs units within the retriever's input limit; normalizes Unicode, whitespace, case, and punctuation; applies a duplicate policy; and aggregates evidence by maximum score, top-three mean, top-two mean, or view-level RRF. The program cannot change retriever parameters, training data, the corpus, relevance labels, or metric code.

Every candidate program is serialized. Evaluation keys use its effective action signature, so syntactically different proposals that compile to the same behavior are counted once. This rule matters for the flat search surface reported in Section 5.

3.3. Frozen retrievers

All retrievers were frozen before their reported evaluations. They span a lexical baseline, a general multilingual embedding model, a patent-specialized model, and two recent general embedding models (Table 1). Context limits are recorded configuration properties; they do not imply equal effective coverage of a patent record. PatEmbed uses CC BY-NC-SA 4.0, which limits commercial deployment. The license is therefore part of the system boundary, not a footnote to the effectiveness result.

Table 1: Frozen retrievers. Short names are used in the text and figures; repository retriever identifiers remain in the provenance files.

Short name	Retriever	Type; dimension; context	License	Role in development
BM25	BM25S 0.3.10	lexical; –; –	–	diagnostic lexical baseline [5]
BGE-M3	BAAI/bge-m3	dense multilingual; 1,024; 8,192	MIT	diagnostic [9]
PatEmbed	datalyes/ patembed-large	patent dense; 1,024; 512	CC BY-NC-SA 4.0	advanced; research/non-commercial
Arctic	Snowflake Arctic-Embed-M-v2.0	dense multilingual; 768; 8,192	Apache 2.0	advanced [10]
Qwen3	Qwen3-Embedding-0.6B	dense multilingual; 1,024; 32,768 declared	Apache 2.0	advanced [11]

4. Experimental design

4.1. Benchmark and evidence stages

Every experiment belongs to the same family-level cross-domain benchmark lineage [1]. The complete retrieval pool contains 45,336 candidate documents. Table 2 records population, exposure, freeze state, and permitted interpretation for each study step. Development is iterative. Selection-125 is the only selection exposure. Final-872 compares exactly two frozen systems. The depth run and candidate-exposure diagnosis use only the confirmed winner after Final-872.

4.2. Development, transfer, and fixed controls

The common screen applies one representation to all five retrievers. The per-retriever search then preserves each recorded decision class (strict improvement, tie, or no strict improvement). The transfer study applies each advanced-retriever program to every advanced target retriever without editing. The 3×3 matrix tests compatibility; it is not a second selection procedure.

Five fixed controls test whether rank-list complementarity changes the result: the strongest single system, top-two RRF, top-three RRF, all-primary fusion, and a commercial-only profile. The all-primary and top-three controls resolve to the same aggregate scores in the canonical artifact. None of

Table 2: Study steps, repository records, populations, and claim boundaries. Repository codes identify experimental records, not retrievers.

Record	Study step	Population	System state and exposure	Permitted inference
A1–A3	Development: common screen, per-retriever search, transfer and controls	development cohorts	five retrievers; iterative search; fixed transfer and fusion controls	describe retriever interaction, transfer, complementarity, ties, and negative results
A4	Selection-125	125 queries; OUT primary $n = 90$	at most four frozen profiles; one exposure	select one profile
A5	Final-872 confirmation	872 queries	two frozen systems; no tuning; paired by query	estimate the confirmatory effect of RCRS against the static comparator
A6	Full-benchmark depth run	1,247 queries	confirmed winner only; depth 200; after Final-872	report depth-wise stress test; no comparator effect
A7	Candidate-exposure diagnosis	strict relation-scoped IN/OUT; OUT has 905 judged queries and 5,193 positive pairs	unchanged winner and its frozen Top-200 pool	diagnose candidate exposure and the within-pool ordering bound

these development controls can replace Selection-125 or enter Final-872 after inspection.

4.3. Selection and confirmation

Selection-125 evaluates the frozen research reference, BALANCED, DEEP, and FAST profiles on 125 queries. The primary OUT subset contains 90 queries. One exposure chooses the research profile. That profile and one static-representation comparator are then frozen for Final-872. Confirmation requires 872/872 coverage and paired query-level comparison; a failure would remain visible in the report.

4.4. Metrics, uncertainty, and population accounting

Recall@k is computed at patent-family level. OUT Recall@100 is primary. nDCG@100 is a confirmatory secondary outcome. nDCG@10 is descriptive because the canonical Final-872 artifact contains no confidence interval for that endpoint. Final-872 differences are paired by query. Percentile 95% confidence intervals use 10,000 paired bootstrap resamples; win, tie, and loss counts use the same query unit.

The label OUT has two denominators that must not be mixed. In Final-872, fields ending in `_out` aggregate the eligible-out Final-872 cohort and contain 17,429 relevant pairs. In the post-confirmatory analyses, strict OUT is relation-scoped and contains 905 judged queries and 5,193 positive pairs. IN and OUT can overlap at query level because one query may have both relation types. Final-872 and the full-benchmark depth run therefore describe different OUT populations; they are not a before–after series.

5. Development results

5.1. Common screen and per-retriever search

Common-screen Recall@100 was 0.1912 for BM25, 0.2699 for BGE-M3, 0.4134 for PatEmbed, 0.3407 for Arctic, and 0.3637 for Qwen3. The corresponding nDCG@100 values were 0.1727, 0.2314, 0.3478, 0.2845, and 0.3079. BM25 and BGE-M3 remained diagnostic; PatEmbed, Arctic, and Qwen3 entered advanced development. Appendix Table A.4 gives the full common-screen and per-retriever aggregate ledger.

The per-retriever search produced different outcomes. Arctic improved from 0.3407 to 0.3587 Recall@100 and met its strict-improvement rule. Qwen3 rose from 0.3637 to 0.3737 but did not meet that rule. PatEmbed reached 0.4230 and retained the recorded classification "numerical tie to common screen." BM25 and BGE-M3 also tied under their decision rules, at 0.2347 and 0.2900. Program utility therefore depended on the frozen retriever.

5.2. Transfer, fusion, and effective search surface

Figure 2A reports all nine advanced-retriever transfers. The three programs produced a narrow Recall@100 range on PatEmbed (0.4184–0.4193), a lower range on Arctic (0.3374–0.3413), and 0.3595–0.3626 on Qwen3. No program source held a consistent diagonal advantage, and the target-retriever ordering remained visible after transfer.

The controls were also non-monotonic (Fig. 2B). Best-single and top-two RRF were nearly tied in Recall@100 (0.4184 and 0.4187), while top-two RRF increased nDCG@100 from 0.3471 to 0.3527. All-primary fusion and top-three RRF both returned 0.4151 Recall@100 and 0.2848 nDCG@10, below the strongest single system on those endpoints. The commercial-only profile returned 0.3693 Recall@100. It quantifies a licensing trade-off within this benchmark; it is not an alternative Final winner.

The HarnessOpt procedure proposed 12 candidates in three batches, but their effective action signatures collapsed to one. Additional proposal rounds did not create additional executable behavior. Nominal proposal count would therefore overstate the amount of search; the effective signature is the reported unit.

6. Selection and confirmatory results

6.1. One selection exposure

All four Selection-125 profiles completed 125/125 queries. The research reference obtained 0.4161 OUT Recall@100. BALANCED and DEEP tied at 0.3606; FAST obtained 0.3083. Against either BALANCED or DEEP, the paired difference was 0.0556 (95% CI 0.0300–0.0817; 49 wins, 20 ties, 21 losses). Against FAST, the difference was 0.1078 (95% CI 0.0811–0.1350; 69/9/12). The canonical Selection-125 artifact contains no verified nDCG result, so none is inferred. The research profile and static comparator were frozen after this exposure.

6.2. Final-872 effectiveness

Both Final-872 systems completed 872/872 queries without failure. RCRS obtained 0.4425 OUT Recall@100; the static system obtained 0.3311 (Fig. 3). The paired difference was 0.1114 (95% CI 0.1023–0.1204), with 619 wins, 158 ties, and 95 losses. nDCG@100 increased from 0.2793 to 0.3656, a paired difference of 0.0863 (95% CI 0.0787–0.0941). Descriptive nDCG@10 increased from 0.2337 to 0.2975, an aggregate difference of 0.0638.

Under the frozen Final-872 protocol, conditional representation search improved the selected system over its static comparator. The estimate applies to this paired comparison; external retrievers and individual grammar components were outside its design.

The operational results were mixed. RCRS reduced median latency and had a similar p95, but its p99 was 1,044.613 ms versus 804.002 ms for the static system. Throughput was higher for RCRS (3.319 versus 2.666 queries per second). The worse tail latency prevents an unqualified efficiency claim.

7. Post-confirmatory diagnosis

7.1. Complete-benchmark depth profile

After Final-872 fixed the winner, the full-benchmark depth run evaluated that system alone on 1,247 queries and 45,336 documents to depth 200. The run produced 249,400 ranked rows with no failure. Overall Recall@10,

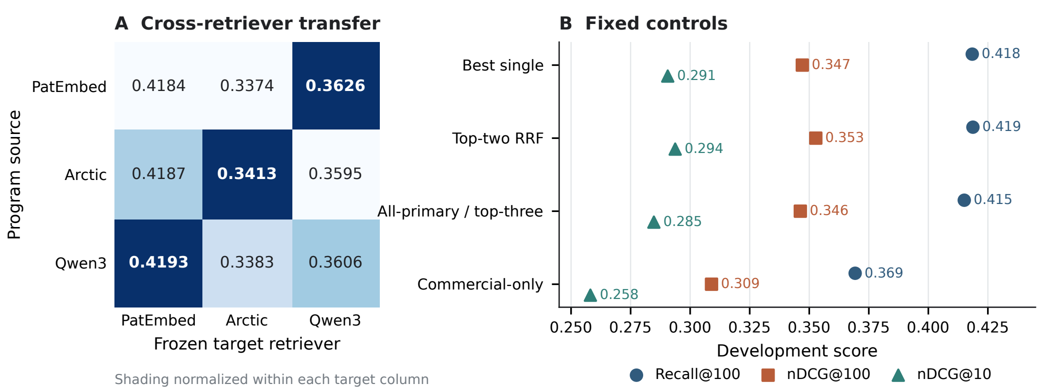

Figure 2: Development-only transfer and fixed controls. (A) Recall@100 for programs optimized on PatEmbed, Arctic, or Qwen3 and applied to each frozen target retriever; shading is normalized within a target column. (B) Fixed fusion and licensing controls. These data were not used to revise the Final-872 systems.

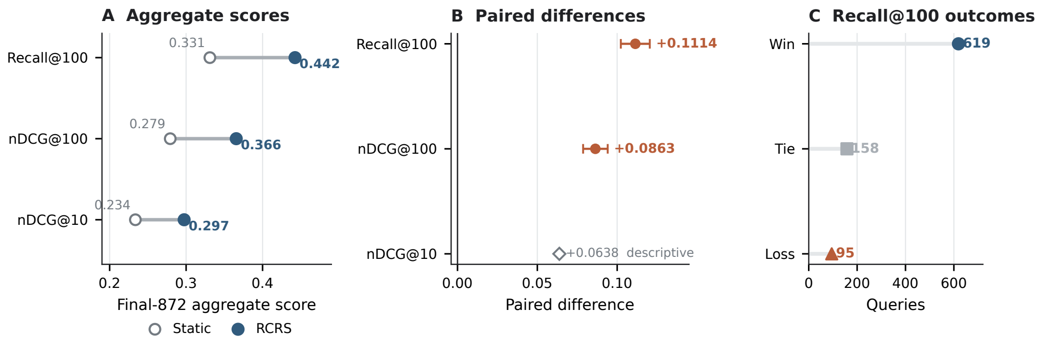

Figure 3: Final-872 comparison of two frozen systems. (A) Aggregate effectiveness. (B) Paired query-level differences and 95% bootstrap confidence intervals; nDCG@10 is descriptive. (C) Query-level Recall@100 wins, ties, and losses.

Table 3: Final-872 effectiveness and operational checks. Confidence intervals are paired query-level bootstrap intervals. The p99 latency result is retained although it favors the comparator.

Outcome	Static	RCRS	Difference or ratio	Statistical or operational note
OUT Recall@100	0.3311	0.4425	+0.1114	CI 0.1023–0.1204; 619/158/95
OUT nDCG@100	0.2793	0.3656	+0.0863	95% CI 0.0787–0.0941
OUT nDCG@10	0.2337	0.2975	+0.0638	descriptive; no canonical interval
Latency p50 / p95 (ms)	315.862 / 735.171	237.547 / 732.218	–	lower median; similar p95
Latency p99 (ms)	804.002	1,044.613	–	higher for RCRS
Throughput / equal-cost ratio	2.666 q/s / –	3.319 q/s / –	1.4568	higher throughput; canonical aggregate cost estimate
Coverage / failures	872/872; 0	872/872; 0	–	both deterministic in the recorded run

@20, @50, @100, and @200 was 0.1368, 0.2147, 0.3364, 0.4390, and 0.5468 (Fig. 4A). IN recall at those depths was 0.1870, 0.2779, 0.4131, 0.5282, and 0.6451. Strict OUT recall was 0.0340, 0.0709, 0.1340, 0.1884, and 0.2602. nDCG@100 was 0.3625 overall, 0.4065 for IN, and 0.0706 for OUT.

The depth run contains no comparator and uses relation-scoped denominators. It describes the frozen winner at increasing candidate depth; it does not re-estimate the Final-872 effect.

7.2. Strict OUT exposure and Top-200 bound

The candidate-exposure diagnosis partitions 5,193 strict OUT-relevant family pairs (Fig. 4B). Of these, 796 appeared in ranks 1–100, 332 only in ranks 101–200, and 4,065 were absent at rank 200. Among 905 judged OUT queries, 67 were fully retrieved by rank 200, 297 were partially retrieved, 86 had relevant evidence only in ranks 101–200, and 455 had no relevant candidate in the Top-200 pool. Appendix Table A.5 records these counts.

The analytical oracle moves each relevant family already present in the frozen Top-200 into the first 100 positions; it adds no candidate. The resulting Recall@100 bounds were 0.5468 overall, 0.6451 for IN, and 0.2602 for OUT, compared with observed values of 0.4390, 0.5282, and 0.1884. The within-pool gaps were 0.1079, 0.1169, and 0.0717 (Fig. 4C). These are ordering bounds, not achieved reranker scores. Even perfect ordering of the exposed Top-200 would leave 4,065 strict OUT-relevant pairs unavailable.

8. Discussion

8.1. Representation is part of the retriever configuration

The development results reject a model-independent account of preprocessing. Arctic met its strict-improvement rule; PatEmbed and Qwen3 did not. Transfer changed scores by target retriever and produced no consistent source-program advantage. These outcomes fit a conditional view: the useful text units and aggregation rule depend on the scoring behavior and input constraints of the retriever that consumes them.

Final-872 shows that this interaction can survive a freeze. The estimate is paired, its interval excludes zero, both systems cover the same 872 queries, and the comparison was fixed after one selection exposure. The negative and tied development outcomes are part of the same result. They show that executable search can find a better configuration without guaranteeing that every retriever, every proposal batch, or every fusion will improve.

8.2. Candidate exposure is a separate engineering target

The exposure decomposition narrows the next question. A reranker can reorder the 332 strict OUT-relevant pairs found only in ranks 101–200 and can change the placement of relevant pairs already in the first 100. It cannot recover the 4,065 pairs absent from the Top-200. Candidate generation, coverage, and cross-domain exposure therefore need separate measurement from ordering quality.

The oracle is reported as a bound. Calling the Top-200 bound a reranking result would imply an implemented model and an attainable gain. Neither exists here. The bound measures only the largest Recall@100 value permitted by the frozen candidate pool.

8.3. Implications for patent-search evaluation

Three reporting practices follow from the evidence. A representation program should be versioned with its retriever, not listed as generic preprocessing. Family normalization, duplicate handling, passage packing, and score aggregation should appear in the executable configuration. Finally, relation-scoped populations should be reported with their query and pair denominators. The same label, such as OUT, is insufficient when protocols use different eligibility rules.

Licensing also constrains system choice. PatEmbed supports the research comparison but its non-commercial share-alike terms may exclude a production deployment. The commercial-only control quantifies one in-benchmark alternative, but it does not establish the best deployable system. A production study would need a new prespecified comparison among models with compatible terms.

8.4. Limitations and follow-up work

The evidence is bounded in six ways. It comes from one family-level benchmark. Final-872 contains one frozen comparator, not an exhaustive baseline suite. The grammar changes several components jointly, and no hash-bound component ablation identifies a causal field, segmentation rule, or aggregation operation. External benchmark configurations were not verified as comparable, so cross-study scores are not ranked. IN and OUT are relation-scoped and can overlap by query. The candidate-exposure diagnosis contains no implemented rescue model or layer-three reranker.

A separate preregistered study could test grammar ablations, additional licensed baselines, an independent benchmark, and candidate-generation methods with a new development/final split. Reusing Final-872 for those decisions would weaken the confirmation reported here.

9. Conclusion

Selecting an executable patent-text program for a frozen retriever increased OUT Recall@100 by 0.1114 over a frozen static representation on

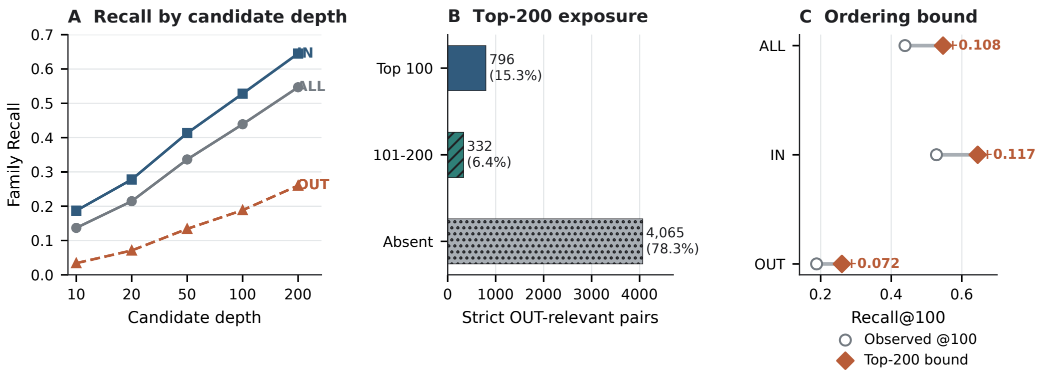

Figure 4: Diagnosis of the unchanged winner after confirmation. (A) Family-level recall by candidate depth. (B) Exposure of 5,193 strict OUT-relevant pairs. (C) Observed Recall@100 and the analytical bound from perfect ordering of the frozen Top-200 pool. IN and OUT are relation-scoped and may overlap by query.

Final-872. The gain is specific to the tested family-level protocol. After confirmation, the unchanged winner exposed only 1,128 of 5,193 strict OUT-relevant pairs in its Top-200, placing most of the remaining error outside the pool available to a reranker. The method result and the exposure limit are separate: RCRS improves representation under a freeze, while broader candidate coverage remains an open retrieval problem.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Data and code availability

The benchmark remains subject to its source terms, and PatEmbed remains subject to CC BY-NC-SA 4.0. The submission package contains aggregate tables, deterministic figure-generation code, frozen configuration identifiers, and provenance paths for every reported number. It does not redistribute query identifiers, patent or family identifiers, relevance judgments, per-query outcomes, or raw rankings. The blinded repository URL, archive identifier, and final availability statement will be added after owner review.

Declaration of generative AI and AI-assisted technologies

During manuscript preparation, the authors used an AI-assisted drafting tool for language and document-structure support. The authors reviewed and edited the output, checked reported results against canonical artifacts, and retain full responsibility for the article. This statement will be reconciled with the journal policy in force on the submission date.

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Appendix A. Aggregate evidence ledger

The appendix is part of the main manuscript. No supplementary scientific file is required. Tables A.4 and A.5 contain aggregate values only.

Table A.4: Common screen (repository A1) and per-retriever search (repository A2). Decision classes reproduce the canonical artifact and are not inferred from displayed decimal changes.

	Common screen			Per-retriever search			
Retriever	R@100	nDCG@100	nDCG@10	R@100	nDCG@100	nDCG@10	Decision class
BM25	0.191200	0.172717	0.160011	0.234667	–	–	diagnostic tie
BGE-M3	0.269933	0.231377	0.198497	0.290000	–	–	diagnostic tie
PatEmbed	0.413400	0.347812	0.289856	0.423000	0.357636	0.299444	numerical tie
Arctic	0.340667	0.284546	0.235538	0.358667	0.301868	0.253229	strict gain
Qwen3	0.363733	0.307930	0.256706	0.373667	0.321262	0.273664	no strict gain

Table A.5: Candidate-exposure diagnosis (repository A7). The strict OUT pair-level total is 5,193; the judged-query total is 905.

Pair-level class	Count	Query-level class	Count
Relevant pair in Top-100	796	fully retrieved by rank 200	67
Relevant pair in ranks 101–200	332	partially retrieved by rank 200	297
Relevant pair absent at rank 200	4,065	deep-only at ranks 101–200	86
		no relevant candidate in Top-200	455